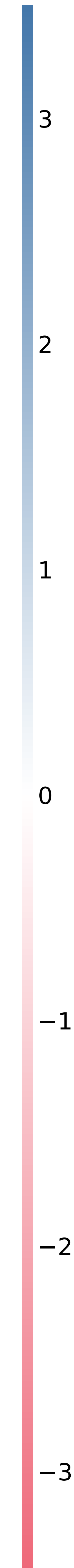
Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:
A. 3.0 amps, 4 ohms
B. 2.8 amps, 3.2 ohms
C. 3.75 amps, 3.2 ohms
D. 4 amps, 3 ohms
E. 3.33 amps, 3.6 ohms
F. 5 amps, 2.4 ohms
G. 4.5 amps, 2.67 ohms
H. 2.2 amps, 5.45 ohms
I. 2.5 amps, 4.8 ohms
J. 3.5 amps, 4 ohms



Reasoning + Answer:

<think>
The problem describes an electric circuit consisting of a battery and a head light bul b connected in series . The battery has a voltage of 12 volts , and the head light bul b has a power diss ipation of 40 watts . We can use the following formula to relate power , voltage , and current :
P = V * I
where :
P = power (40 watts)
V = voltage (12 volts)
I = current (which we want to find)

R ear r anging the formula to solve for current :
I = P / V
= 40 / 12
= 3 . 33 amps

Next , we need to find the resistance of the head light bul b . We can use Oh m 's law :
V = I * R
where :
V = voltage (12 volts)
I = current (3 . 33 amps)
R = resistance (which we want to find)

R ear r anging the formula to solve for resistance :
R = V / I
= 12 / 3 . 33
= 3 . 6 oh ms

Therefore , the current drawn by the head light bul b is 3 . 33 amps , and its resistance is 3 . 6 oh ms .
</think><answer> E . 3 . 33 amps , 3 . 6 oh ms </answer>